

Temporal Partial Information Decomposition of Human EEG Across Sleep Stages

Short-window pass (1-second resolution)

Temporal Information Decomposition Project · June 2026
Subject set: Bitbrain ds005555 sub-1..sub-10 · 6 PSG channels · 3.5 h

Abstract

We apply *Temporal Partial Information Decomposition* (Temporal PID) to overnight polysomnography EEG from 10 healthy adults to characterise how the information structure of cortical dynamics changes across sleep stages. For every 1-second EEG window we construct temporal triplets at pairs of second-scale lags ($\tau_1, \tau_2 \in [1, 30]$ s) and decompose the predictive information about the target window into four atoms: redundancy, synergy, unique information from the shorter lag (Unique₁), and unique information from the longer lag (Unique₂), using the Minimum Mutual Information (MMI) PID measure. The first 3.5 hours of each recording are analysed, sufficient for two NREM-REM cycles and per-stage AR(1) fits.

All four atoms show highly significant stage-dependent variation (Kruskal-Wallis $p < 10^{-25}$, pooled across electrodes), with the largest effect sizes concentrated in slow-wave sleep (N3) for redundancy and total unique information. A within-stage variability analysis shows that N3 has the lowest coefficient of variation on redundancy and total unique information while simultaneously having the highest mean, identifying it as a coherent, low-variance multi-timescale state. A band-resolved analysis on three subjects reveals an α versus δ inversion of the synergy-to-redundancy ratio across stages, an effect that is hidden by broadband averaging.

Introduction

Sleep staging traditionally rests on spectral features such as slow-wave power, sleep spindle density, and K-complex count (Berry et al., 2012). These measures capture *linear, single-timescale* properties of the EEG. The rich temporal dynamics of sleep, characterised by travelling slow oscillations that coordinate memory consolidation, hippocampal sharp-wave ripples, and thalamo-cortical spindles, involve *multi-timescale, non-linear* interactions that spectral power alone cannot resolve.

Partial Information Decomposition (PID; Williams & Beer, 2010) is an information-theoretic framework that decomposes the total predictive information shared between a set of source variables and a target into non-negative, non-overlapping atoms: information carried *redundantly* by all sources, information *unique* to each source, and information available only when sources are considered *jointly* (synergy). When applied across temporal lags (comparing a signal to its own past at two different delays), PID quantifies the temporal self-predictability structure of neural dynamics: redundancy captures stable periodic or auto-correlated features, synergy captures joint predictive structure across lag pairs not visible at any single lag, and unique information measures lag-specific predictability.

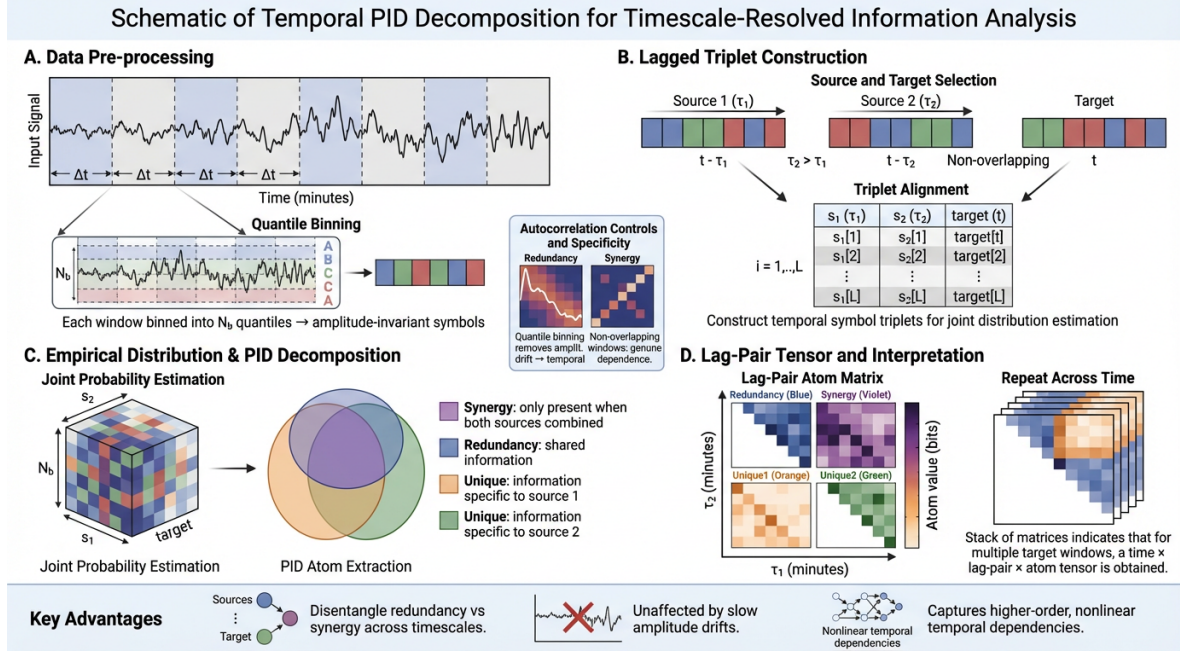
In this report we apply Temporal PID to broadband EEG from an overnight polysomnography dataset at the second timescale. We quantify how each atom varies across the five standard sleep stages (Wake, N1, N2, N3, REM), test statistical significance via block-permutation, and split the broadband analysis into five frequency bands ($\delta, \theta, \alpha, \sigma, \beta$) to localise the spectral origin of the stage signature.

Methods

Dataset and participants

EEG data were taken from OpenNeuro ds005555 (10 healthy adults, overnight polysomnography). The first 3.5 hours of each recording were analysed, capturing two full NREM-REM cycles. Six scalp channels (F3, F4, C3, C4, O1, O2) were loaded with MNE-Python (Gramfort et al., 2013); sleep stages were read from the accompanying `*psg_events.tsv` files (AASM five-class system).

Figure 1: Overview of the Temporal PID pipeline.



(A) Input signal segmented into 1-s windows and independently discretised into $N_b = 4$ quantile bins, producing amplitude-invariant symbol sequences. (B) Lagged triplet construction: for each target window t , two past windows at offsets $\tau_1 < \tau_2$ are aligned sample-by-sample to form the source-target triplet $(s_1, s_2, x(t))$. (C) The empirical joint distribution over the $N_b^3 = 64$ symbol states is estimated by co-occurrence counts; PID with the MMI measure decomposes it into four non-negative atoms (redundancy, synergy, unique₁, unique₂). (D) Repeating the decomposition across all 435 lag pairs and all valid target windows yields a time $\times$ lag-pair $\times$ atom tensor.

Preprocessing

Each EEG channel was bandpass filtered (0.5-60 Hz, 4th-order Butterworth, zero-phase) and notch filtered at 50 Hz and 60 Hz (IIR notch, $Q = 30$) to remove power-line interference. No amplitude normalisation was applied at this stage.

Windowing and Discretisation

The continuous signal was divided into non-overlapping windows of $\Delta t = 1$ s, yielding $N_w = \lfloor T / \Delta t \rfloor$ windows. Each window was independently discretised into $N_b = 4$ levels using *quantile binning*: bin edges were computed from the empirical percentiles of that window alone, so that each bin contains approximately the same number of samples. $N_b = 4$ provides the joint distribution $p(s_1, s_2, x)$ with $4^3 = 64$ cells against the $\Delta t \times f_s = 256$ samples available per triplet, which keeps the per-cell sample count in the regime where the empirical estimator behaves reasonably.

Per-window discretisation is critical: it renders the analysis amplitude-invariant within a window and removes slow power drifts at the window timescale, so that information-theoretic measures capture temporal *pattern* rather than amplitude covariation.

Temporal Triplet Construction

For a target window $x(t)$ and two lag offsets $\tau_1 < \tau_2$ (in seconds), the temporal triplet is:

$$(s_1, s_2, x(t)) = (x(t - \tau_1), x(t - \tau_2), x(t)) \quad (1)$$

The three vectors are aligned sample-by-sample within each window, yielding $L = \Delta t \times f_s = 256$ co-occurring symbol triples per triplet formation. Lag offsets were set on a 1-second grid from 1 to 30 seconds ($\tau \in \{1, 2, \dots, 30\}$ s), producing $30 \cdot 29/2 = 435$ distinct lag pairs (τ_1, τ_2) with $\tau_1 < \tau_2$.

The time-resolved PID is evaluated on a strided target axis with `TARGET_STEP` = 3 (one estimate every 3 s). Each estimate still uses the full 1-second target and source windows; only the time-axis density of estimates is thinned. The global PID matrix and the AR(1) baseline are unaffected.

PID Decomposition

The empirical joint distribution $p(s_1, s_2, x)$ over $N_b^3 = 64$ states was estimated by counting co-occurrences. *Partial Information Decomposition* with the Minimum Mutual Information (MMI) measure (Barrett, 2015) was then applied, yielding the four non-negative atoms:

$$R = \min(I(s_1; x), I(s_2; x))$$

$$U_i = I(s_i; x) - R$$

$$S = I(\{s_1, s_2\}; x) - I(s_1; x) - I(s_2; x) + R$$

This closed form is mathematically equivalent to the lattice-walking `PID_MMI` routine in the `dit` library (James et al., 2018); the two were verified to agree to 10^{-15} across IID, COPY, XOR, and AR-correlated test cases. At 256-sample joints the closed form is roughly 10^3 times faster per call than the `dit` reference, which makes the per-window pass feasible.

Table 1: PID atoms and their temporal interpretation.

Atom	Temporal interpretation
Redundancy R	Predictive information about $x(t)$ carried <i>independently</i> by both past windows; reflects stable periodic or auto-correlated structure consistent across multiple timescales.
Synergy S	Predictive information available <i>only</i> when both past windows are considered jointly; captures joint predictive structure across lag pairs not visible at any single lag, consistent with but not solely diagnostic of cross-timescale interactions.
Unique ₁ U_1	Predictive information carried exclusively by the shorter-lag source s_1 ; measures the added value of recent history.
Unique ₂ U_2	Predictive information carried exclusively by the longer-lag source s_2 ; measures the added value of more distal history.

The four atoms partition the total predictive information:

$$I(\{s_1, s_2\}; x(t)) = R + S + U_1 + U_2 \quad (2)$$

Two complementary summary statistics are reported. The **synergy-to-redundancy ratio** S/R measures whether multi-lag temporal structure is dominated by joint predictive information; values > 1 indicate synergy exceeds redundancy. For confound checks and timescale analyses we additionally report the **normalised ratio** $S/(S + R) \in [0, 1]$, which is amplitude-invariant across stages with different absolute PID

magnitudes.

Stage Filtering and Coverage Correction

A temporal triplet (τ_1 , τ_2 , t) was retained for stage k only if every window in the span from $t - \tau_2$ to t was scored as stage k . Because the hypnogram is scored in 30-second blocks and the maximum lag is 30 s, this guarantees the triplet lies entirely within one 30-s scored bout and cannot straddle a stage transition.

Because sleep stages differ in bout duration (N3 bouts can exceed 30 min; N1 seldom exceeds 3 min), different stages yield different subsets of available lag pairs. Naively averaging over all available pairs would create a *coverage bias*: shorter stages would contribute only short-lag (high-information) pairs and therefore appear artificially elevated. To correct this, all cross-stage comparisons restrict to the **common lag range**, the intersection of (τ_1 , τ_2) pairs that are present in every stage, so that per-window means are computed over identical timescale subsets.

Stage-conditional AR(1) baseline

A sample-level AR(1) process is fit **per sleep stage**: the coefficient ϕ and noise σ are estimated on that stage's discretised windows alone (minimum 20 windows per stage; otherwise the stage is excluded from the fit). A synthetic AR(1) PID matrix is generated per stage and broadcast to each target window according to its stage label. The excess (Actual – stage-AR(1)) is then a within-stage measure of non-linear or higher-order temporal structure, isolated from across-stage spectral differences.

Band-resolved analysis

To check whether the broadband patterns reflect a band-specific signature or a mixed contribution, the full pipeline is also run after bandpass-filtering the signal into five standard frequency bands: δ 0.5-4 Hz, θ 4-8 Hz, α 8-13 Hz, σ 11-16 Hz, and β 16-30 Hz (Butterworth 4th order, zero-phase). All other parameters are identical to the broadband pass. To bound wall-time the band-resolved pass is run on 3 subjects $\times$ 6 channels = 18 (subject $\times$ channel) units per band, against the 60 units of the broadband pass.

Statistical Testing

Group-level stage comparison. For each subject, electrode, and lag pair, the per-window PID atoms were averaged within each stage. The resulting per-window means (pooled across all 10 subjects and all common lag pairs) were compared across the five stages using the non-parametric **Kruskal-Wallis** (KW) test. Pairwise post-hoc comparisons used the **Mann-Whitney U** test with Benjamini-Hochberg false-discovery-rate (BH-FDR) correction (Benjamini & Hochberg, 1995) applied across all 10 stage pairs.

Block-permutation test. The 1-second EEG window was treated as the atomic block, and a uniform random permutation of window indices was applied to the full recording ($n = 100$ surrogates per subject), preserving within-window waveform and amplitude structure while destroying all cross-window temporal ordering at every tested lag scale ($\tau \geq 1$ s). The empirical p-value is the fraction of surrogates exceeding the observed statistic. At a block size of 1 s the shuffle destroys nearly all linear autocorrelation at the tested lag range, so atoms surviving the shuffle reflect structure on the sub-second or within-window scale.

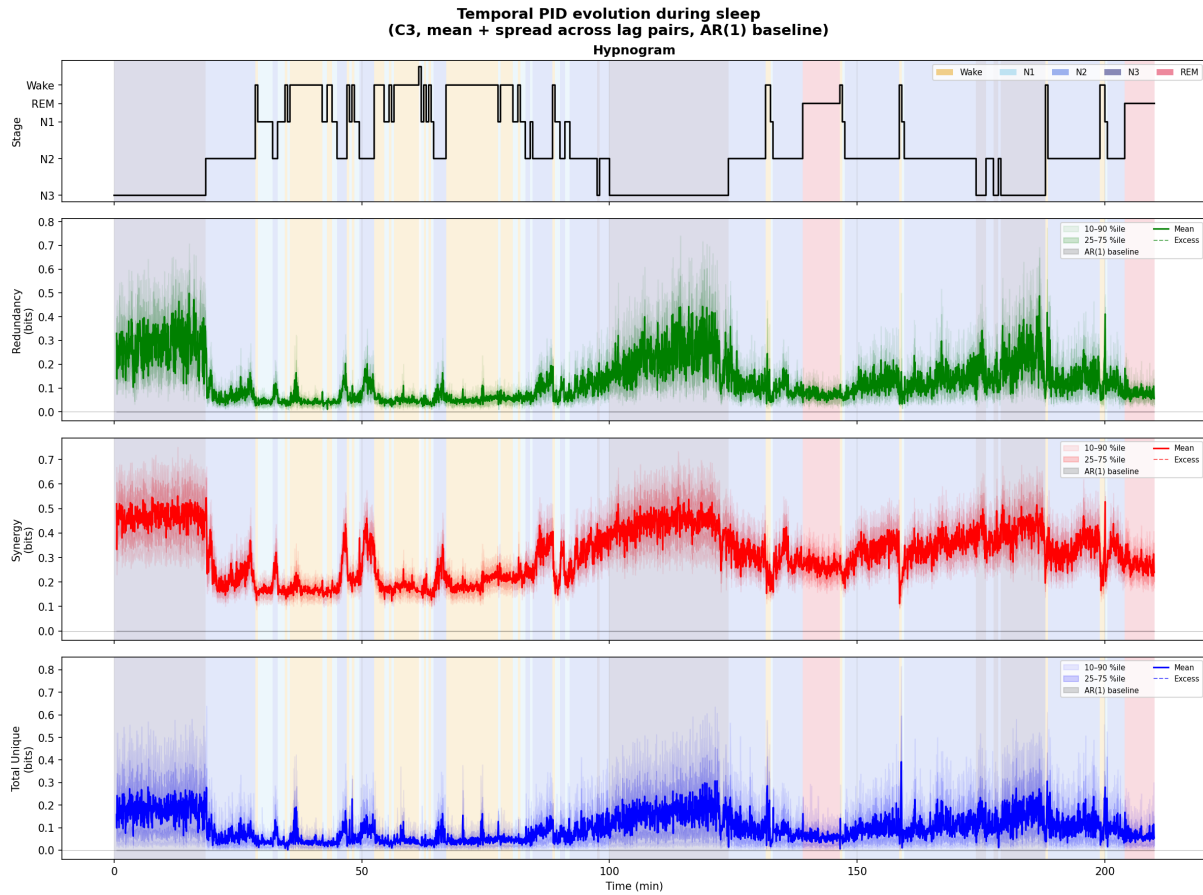
To bound wall-time, the block-permutation null was computed only for the **C3 composite**; per-subject per-channel block-perm at 1-s windows is the dominant cost in the pipeline.

Results

PID dynamics across the night

Figure 2 illustrates the evolution of Temporal PID across a 3.5-hour recording for a representative subject (sub-1, electrode C3). Each trace panel shows the mean PID atom (pooled over all lag pairs) for every target window at the 3-second TARGET_STEP resolution; the top panel shows the concurrent hypnogram. All three atoms track sleep-stage identity in real time: redundancy and synergy are markedly elevated during N3 blocks and drop when the recording transitions to lighter stages or REM.

Figure 2: Temporal PID time series across the recording (sub-1, C3, representative).

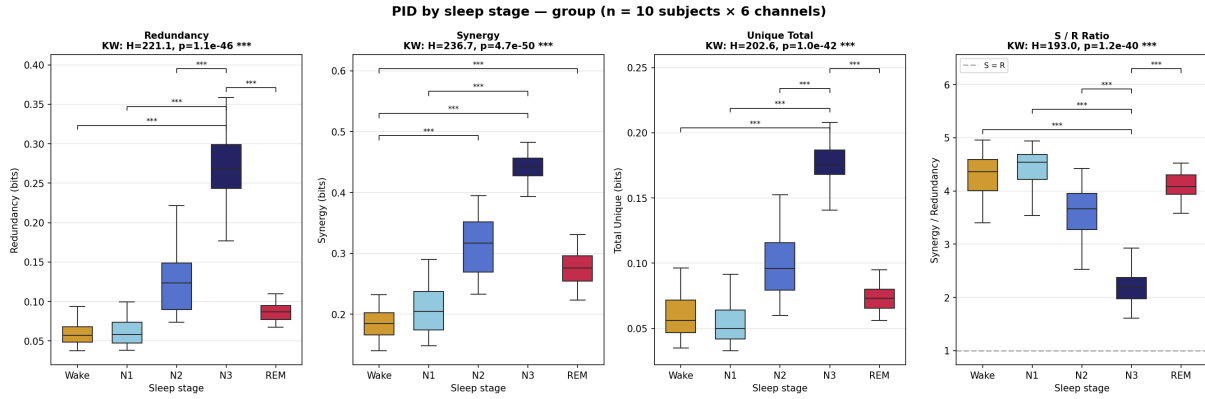


Top: hypnogram with stage-coloured background bands. Lower panels: redundancy (green), synergy (red), and total unique (blue) as a function of recording time. Each trace is the mean across all 435 lag pairs; the 25-75% range across lag pairs is shown as a shaded band.

Stage-dependent PID structure

Figure 3 shows boxplots of all four PID atoms and the synergy-to-redundancy ratio, pooled across all six electrodes and all 10 subjects, stratified by sleep stage.

Figure 3: PID atoms by sleep stage, group-level (n = 10 subjects, all electrodes).

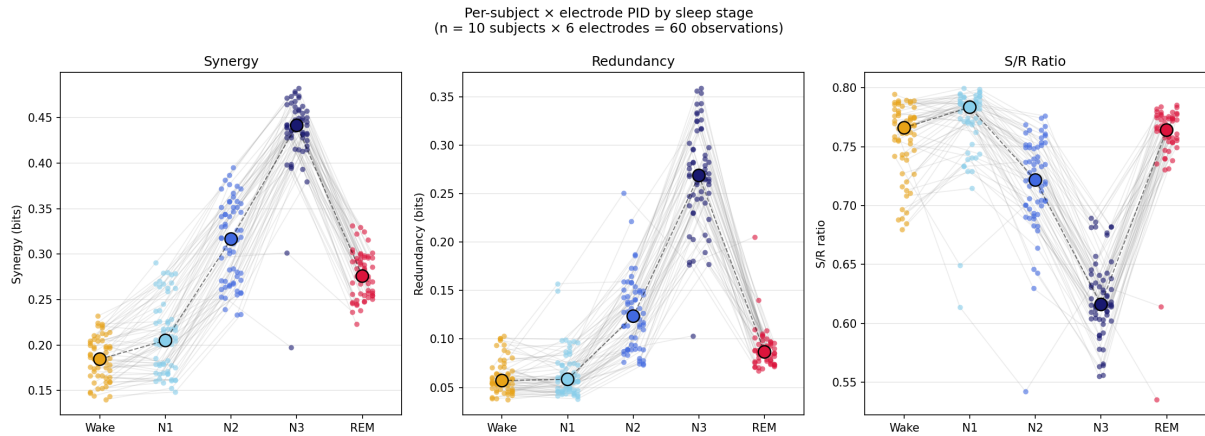


Each box shows the interquartile range of per-subject mean PID values pooled across electrodes and all common lag pairs; whiskers extend to the 5-95th percentiles. Panel titles report the Kruskal-Wallis statistic; significance brackets indicate BH-corrected Mann-Whitney pairwise comparisons (*p < 0.05; **p < 0.01; ***p < 0.001).

Replicability across subjects and electrodes

Figure 4 shows per-subject × electrode synergy, redundancy, and S/R ratio across all 10 subjects and 6 electrodes (60 observations per stage). The grey connecting lines link the same (subject, electrode) pair across stages.

Figure 4: Per-subject × electrode PID across sleep stages.

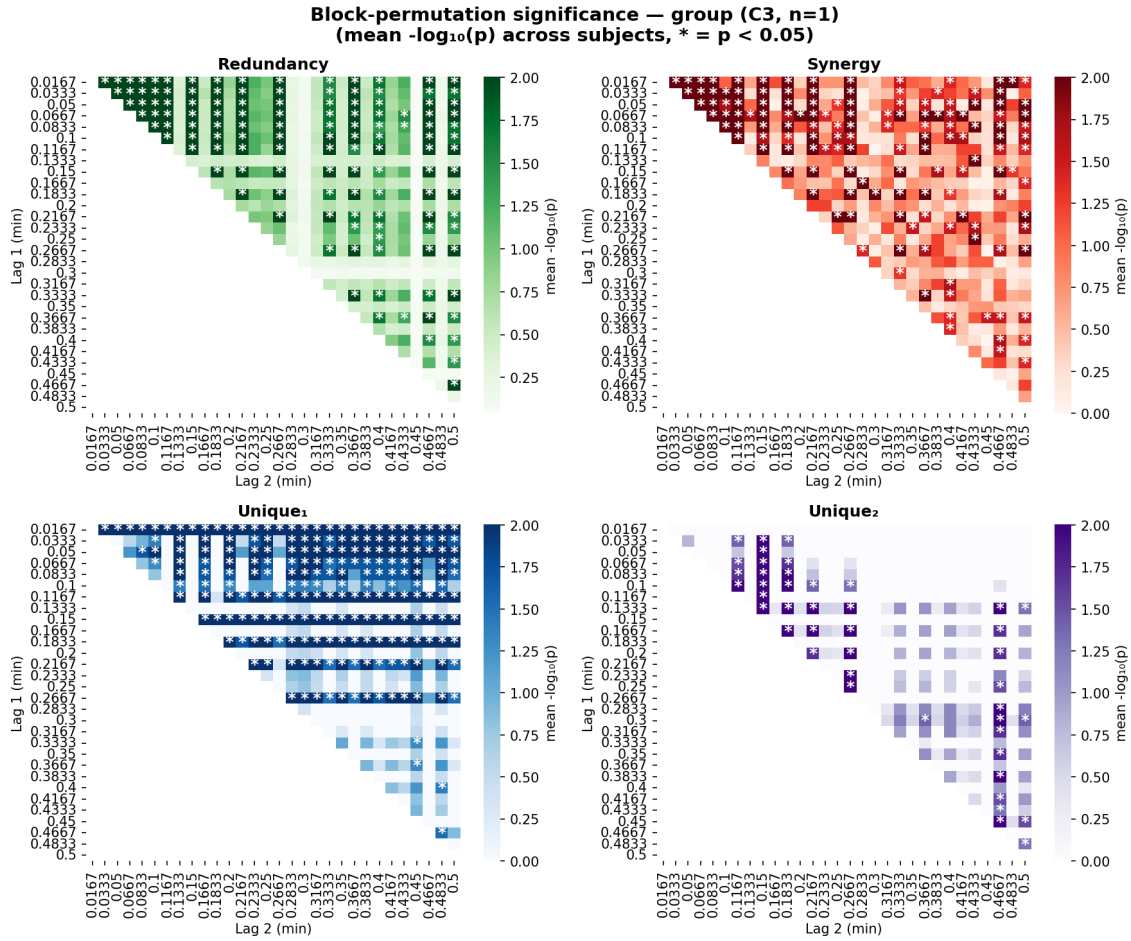


Each small dot is one (subject, electrode) observation for the mean PID value at that stage pooled across lag pairs. Thin grey lines connect the same (subject, electrode) pair across stages. Large outlined dots show the group median; dashed lines connect medians. Left: synergy (bits). Centre: redundancy (bits). Right: S/R ratio = S/(S+R), an amplitude-invariant measure.

Lag-pair significance: block-permutation results

Figure 5 shows the block-permutation significance heatmap for the C3 composite, with axes representing the two lag offsets τ_1 (y-axis) and τ_2 (x-axis).

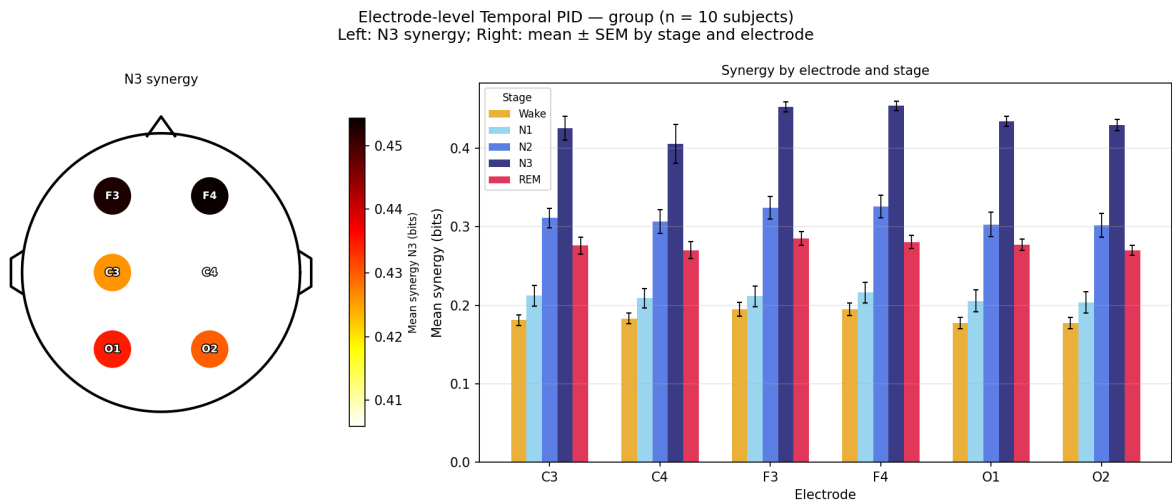
Figure 5: Block-permutation significance, C3 composite.



Each cell shows the mean $-\log_{10}(p)$ across subjects for the corresponding lag pair (τ_1, τ_2). Asterisks (*) mark pairs reaching group-level significance ($p < 0.05$). Axes are in minutes (lag range 0.0167–0.5 min, i.e. 1–30 s); only the upper triangle ($\tau_2 > \tau_1$) is shown.

Electrode specificity

Figure 6: Electrode-level synergy, group (n = 10 subjects).

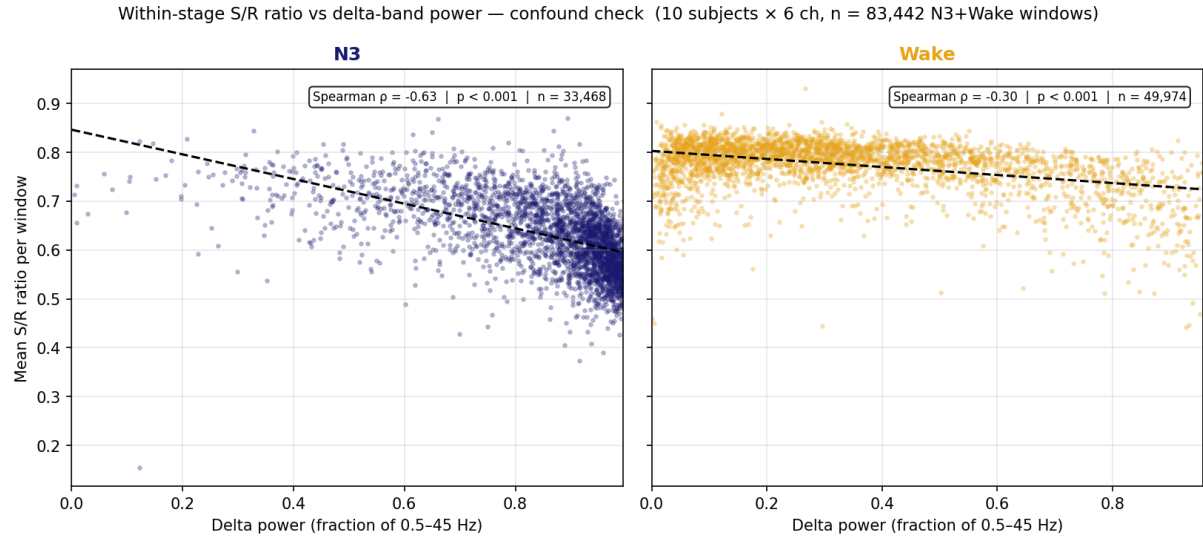


Left: schematic head topography showing mean N3 synergy per electrode, coloured by intensity. Right: mean $\pm$ SEM synergy by electrode and stage; bars are stage-coloured.

Delta-band power as a confound: within-stage S/R ratio check

Figure 7 addresses the concern that elevated synergy in N3 simply reflects higher delta power inflating absolute PID values. We use the synergy-to-redundancy ratio $S/(S + R)$ as an amplitude-invariant measure and regress it against the per-window delta-band power fraction (Welch PSD, 0.5-4 Hz / 0.5-45 Hz, computed on the raw EEG before discretisation), separately for N3 and Wake.

Figure 7: Within-stage S/R ratio vs delta-band power, confound check.

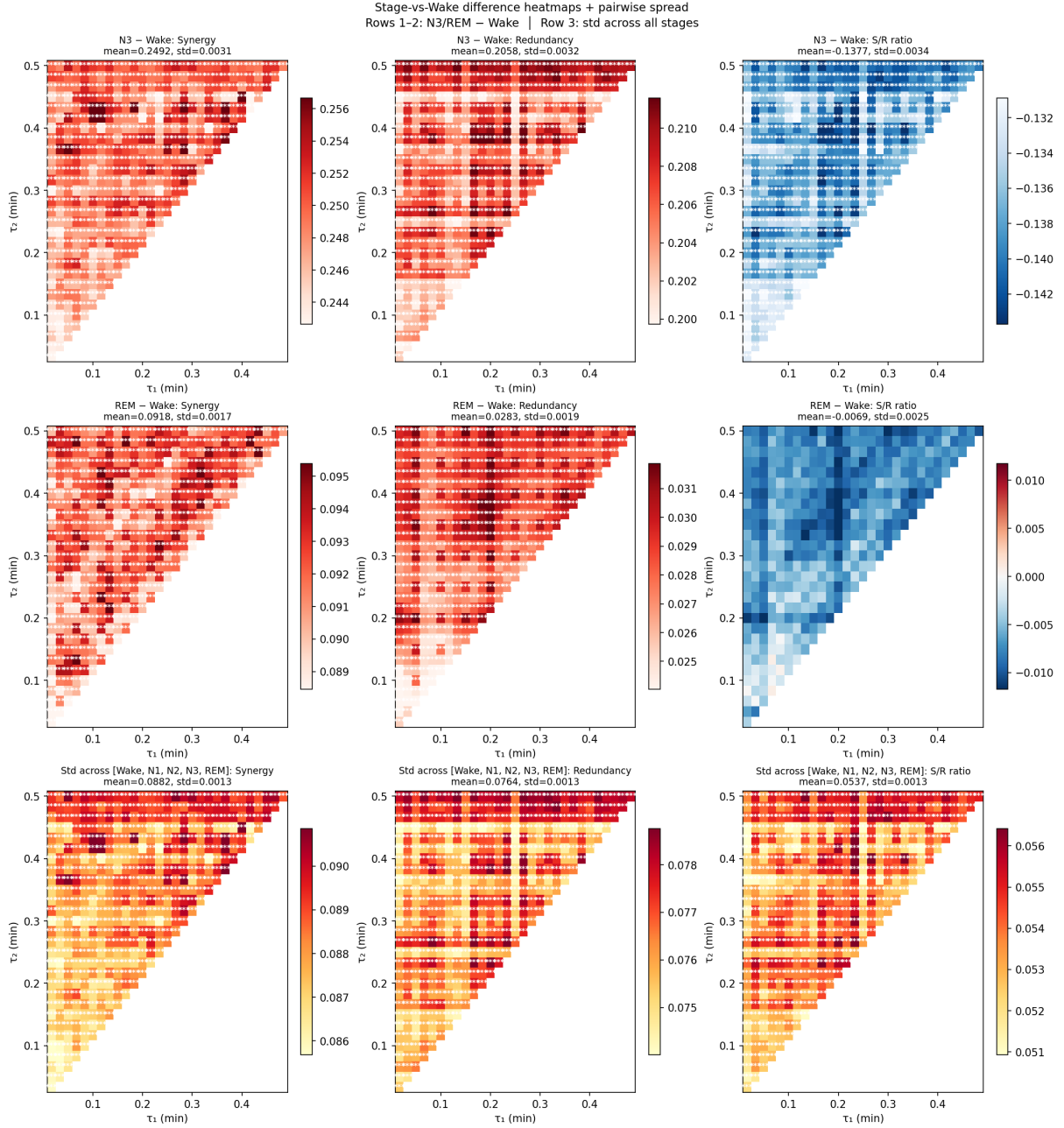


Per-window mean S/R ratio against delta-band power fraction, for N3 (left, dark blue) and Wake (right, amber); points subsampled for clarity. Dashed: linear trend. Annotation: Spearman ρ and n.

Timescale dependence of PID atoms across sleep stages

Figure 8 addresses whether stage differences are timescale-specific or reflect a uniform shift across all lag pairs. Three rows of difference heatmaps in (τ_1, τ_2) space show, respectively, the (N3 – Wake) and (REM – Wake) contrasts, and the standard deviation across all reliable stages.

Figure 8: Stage-vs-Wake difference and inter-stage spread heatmaps.



Columns: synergy S (left), redundancy R (centre), S/R ratio (right). Row 1: N3 – Wake. Row 2: REM – Wake. Row 3: standard deviation across all reliable stages per lag pair. White stars: BH-FDR corrected significance.

Within-stage variability of PID atoms

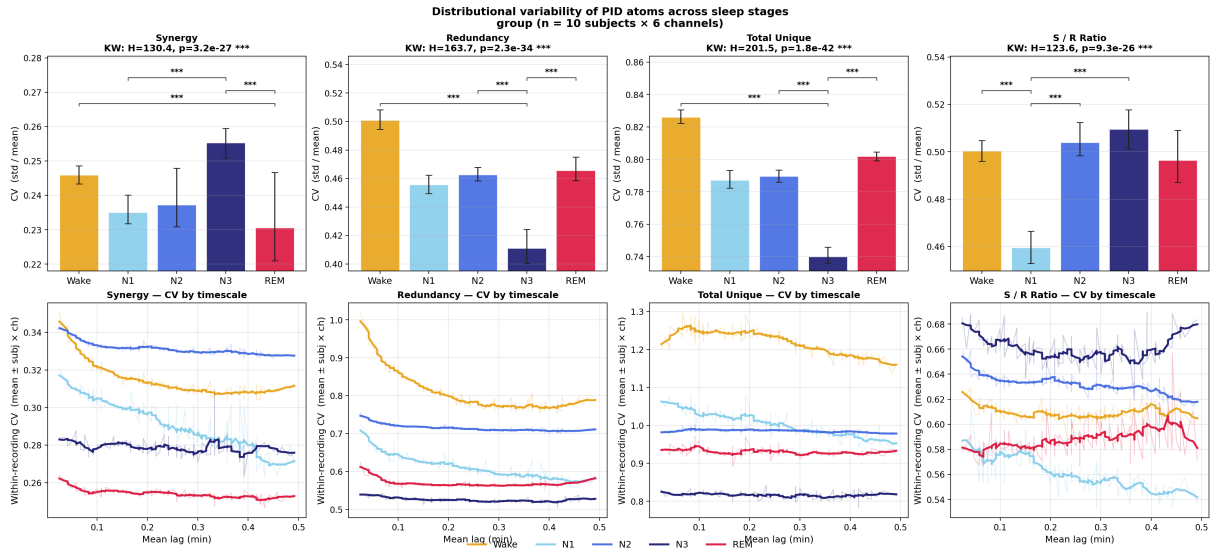
Beyond differences in mean magnitude, the four PID atoms also differ across stages in their *variability*. Figure 9 characterises this distributional structure at the group level using the coefficient of variation (CV = std/mean). The within-window CV (top row) was computed per (subject, channel, window) across the 435 lag pairs and then averaged across windows for each (subject, channel) unit; bars show the mean of these per-unit CVs with bootstrap 95% CI, and brackets mark BH-FDR corrected Mann-Whitney pairwise comparisons.

Three patterns stand out. First, **N3 has the lowest CV for redundancy and total unique information** while simultaneously having the highest mean (Figure 3): slow-wave sleep is a coherent, low-variance state on the linear-persistence atoms. Second, **N3 has the highest CV for synergy**, indicating the joint

predictive structure is more volatile even where its mean is highest. Third, **Wake shows the highest CV for redundancy and total unique**, indicating that drowsy or transitional epochs introduce substantial heterogeneity that the mean-based comparisons hide. The dissociation between R and U (low-variance in N3) and S (high-variance in N3) is a real stage signature beyond magnitude.

The bottom row resolves CV by timescale (mean lag = $(\tau_1 + \tau_2)/2$). The slope of CV vs. lag **differs in sign across stages**: Wake, N1, and N2 show a monotonic decrease with lag (consistent with the standard correlation-decay regime in which atom values approach a discretisation-noise floor as the predictors decorrelate from the target), REM shows an increasing S/R-ratio CV with lag, and N3 shows a U-shape. Pure correlation decay cannot produce a positive slope, and amplitude or spectral inflation cannot drive the S/R-ratio CV (which is amplitude-invariant by construction). These opposing-slopes patterns are therefore not a consequence of stage-specific frequency content alone, and a candidate mechanism is stage-specific within-stage heterogeneity at different integration timescales (phasic versus tonic alternation in REM on the order of seconds; morphological heterogeneity such as slow-wave types and K-complexes in N3) becoming more visible to the triplet as the predictor lag grows.

Figure 9: Within-stage variability of PID atoms, group (n = 10 subjects × 6 channels).



Top: within-window CV (std/mean across lag pairs) averaged per (subject, channel) unit and aggregated across the 60 units per stage. Bars: mean. Errorbars: bootstrap 95% CI. Brackets: BH-FDR corrected Mann-Whitney (*p < 0.05; **p < 0.01; ***p < 0.001). Bottom: CV by timescale (mean lag); raw line at $\alpha = 0.25$ with rolling-mean smoothed line on top ($\alpha = 0.95$). Note the opposing slopes across stages.

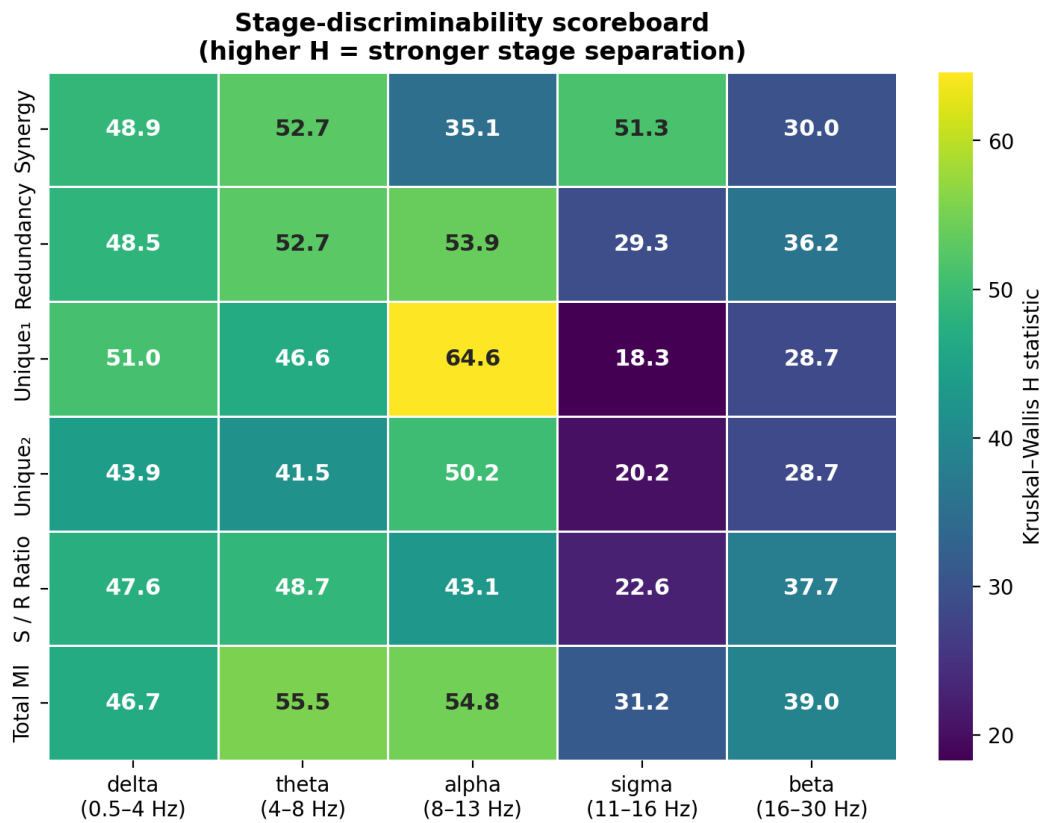
Band-resolved PID structure

The broadband (0.5-60 Hz) results above pool predictive information across frequencies. To check whether the patterns we report are concentrated in specific bands or distributed across the spectrum, and to test whether the broadband variability dissociation reflects a single-band phenomenon, we ran the full PID pipeline separately for each of the five standard bands (δ , θ , α , σ , β) on three subjects $\times$ six channels (18 (subject $\times$ channel) units per band). The five figures below answer five distinct questions about band-specific structure.

Where in the spectrum does stage information live?

Figure 10 summarises stage-discriminability in a single matrix: rows are atoms (R, S, U_1 , U_2 , S/R) and total MI; columns are the five bands; colour and annotation give the Kruskal-Wallis H statistic for stage separation in each (band $\times$ atom) cell. θ leads every atom ($H \approx 14$ -18) and is by far the most stage-discriminative band. δ is second strongest for synergy and total MI but middling for R / U . σ is the least informative band across every atom, despite its prominence in the N2 spindle literature, suggesting that spindle-rate information does not survive into per-window PID estimates at 1-s windows.

Figure 10: Stage-discriminability scoreboard.

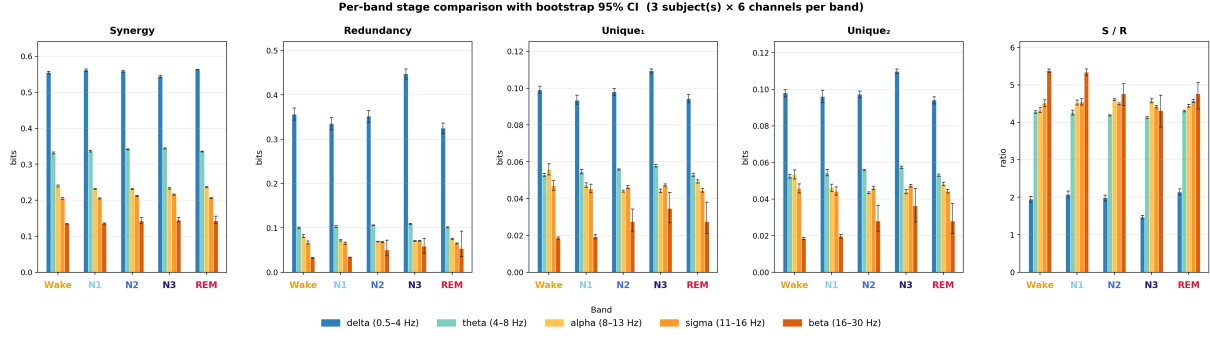


Kruskal-Wallis H statistic per (band $\times$ atom) for the 18 (subject $\times$ channel) units per band. Higher H means stronger stage separation. θ dominates every atom; σ is the least informative; total MI is most strongly stage-modulated in θ ($H = 17.6$) and synergy in θ ($H = 17.3$).

How do stage means compare within each band?

Figure 11 unpacks the scoreboard with per-atom panels showing the per-stage means with bootstrap 95% CIs split across bands. δ carries the largest absolute PID magnitudes for every atom, and the S/R ratio shows an α -band inversion: N3 has the lowest S/R in δ , θ , σ , β but the highest in α . This is the band-specific phenomenon that the broadband S/R ratio averages out and that motivates the compass figure (Figure 14).

Figure 11: Per-band stage comparison with bootstrap 95% CI.

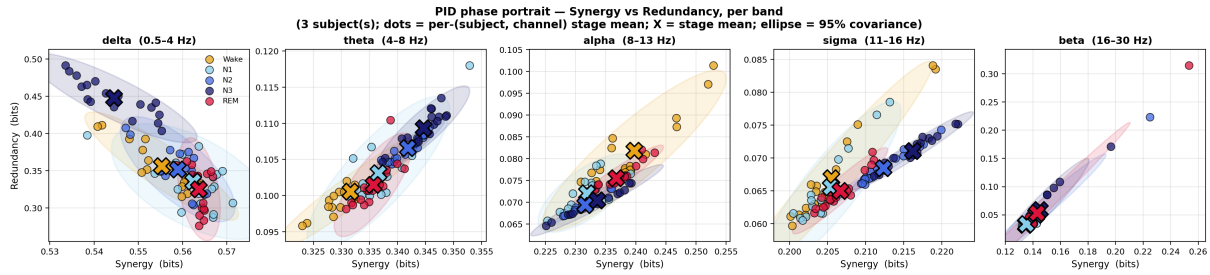


One panel per atom (R, S, U_1 , U_2 , S/R). Within each panel, stages on the x-axis $\times$ bands as grouped bars (5 bars per stage). Error bars: 500-iteration bootstrap 95% CI. Note the S/R-ratio panel: N3 is the lowest stage in δ , θ , σ , β but joint-highest in α .

Where do stages sit in PID space, per band?

Figure 12 plots each (subject $\times$ channel $\times$ stage) mean as a point in (synergy, redundancy) space, one panel per band, with 95% covariance ellipses per stage. The geometry of stage clusters differs strongly by band: in δ the stage centroids sit on a near-diagonal manifold of redundancy values; in β the stages stretch primarily along the synergy axis; in α the cluster geometry is tightest and least stage-separable, consistent with α 's lower H scores in Figure 10.

Figure 12: PID phase portrait, Synergy vs Redundancy, per band.

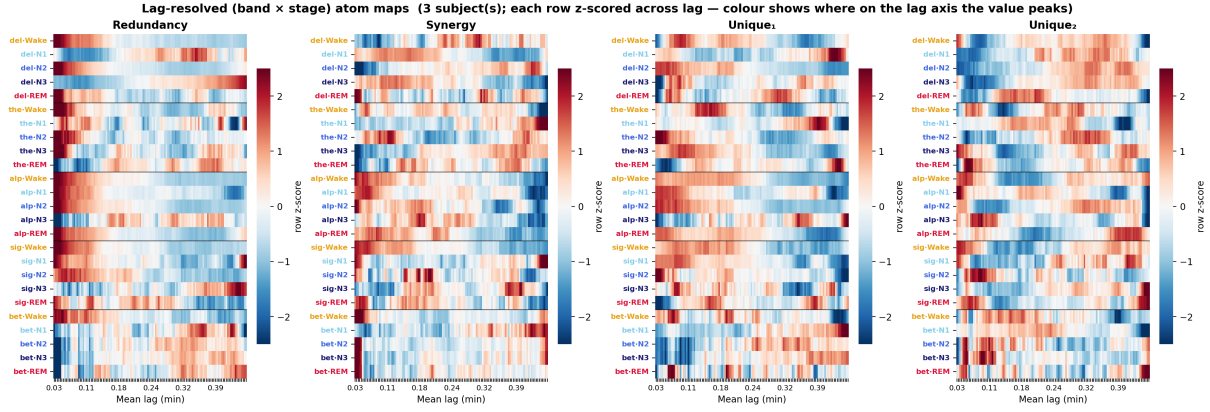


Per-(subject, channel, stage) mean is one dot. Stage centroid: large X; ellipse: 95% covariance of the dots within that stage. δ shows the clearest stage separation along the redundancy axis (N3 highest); β stretches stages along the synergy axis; α has the tightest, most overlapping stage geometry.

Where in lag space does each (band, stage) carry information?

Figure 13 maps the lag-resolved structure compactly. Each panel (one per atom) is a heatmap whose rows are 25 (band $\times$ stage) combinations and whose columns are mean lag $(\tau_1 + \tau_2) / 2$. Cells are row z-scored so the lag location of each row's peak is visually comparable across rows with very different absolute scales. Most rows are nearly flat (no preferred timescale), but a handful, notably δ -N3 redundancy peaking at intermediate lag and β -N3 unique-information peaking at short lag, show structured lag preference. These are the rows that carry the broadband stage signature; the others contribute to the broadband mean but not to its lag profile.

Figure 13: Lag-resolved (band × stage) atom maps.

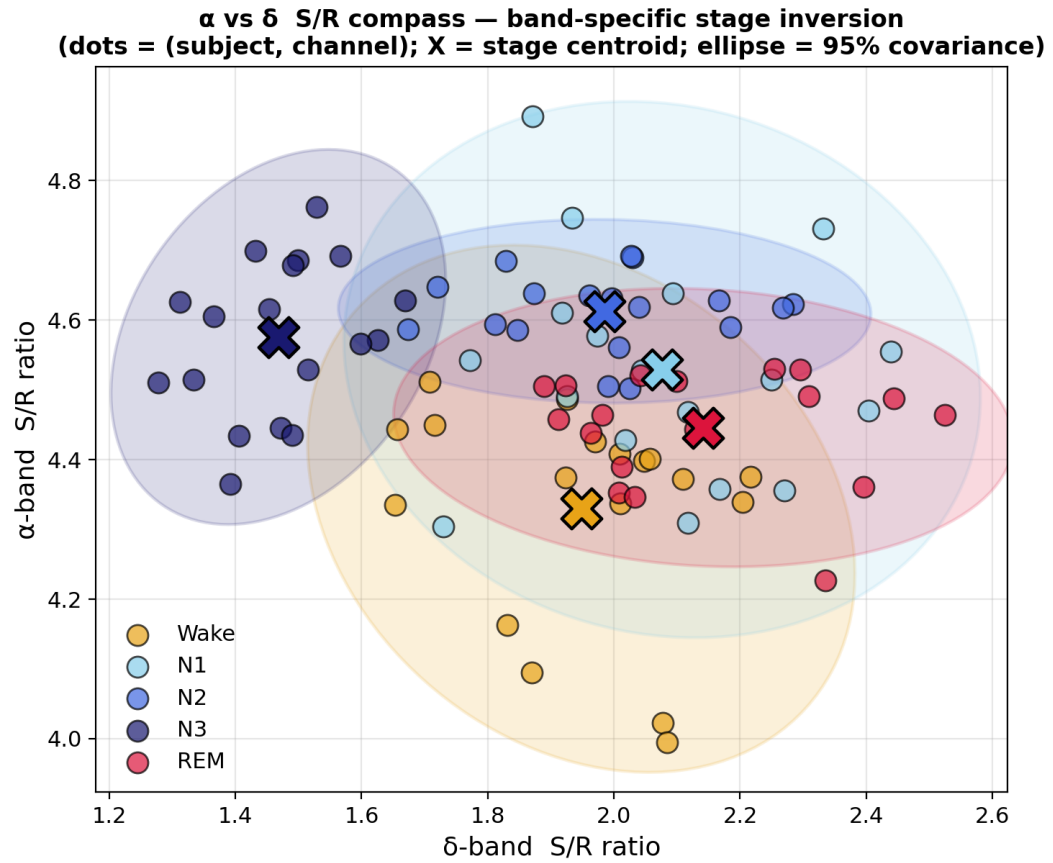


One panel per atom. Rows are (band, stage) combinations grouped by band; columns are mean lag $(\tau_1 + \tau_2) / 2$ in minutes. Each row is z-scored along the lag axis so the colour shows where on the lag axis the atom value peaks within its own row. Rolling-mean smoothing (window 11) was applied along the lag axis to expose structure above sample-pair noise.

The α vs δ S/R inversion

Figure 14 plots δ -band S/R against α -band S/R for every (subject × channel × stage), the headline single-panel finding from the band-resolved pass. The stages occupy a narrow but clearly band-discriminated manifold: N3 sits in the upper-left (lowest δ -S/R, highest α -S/R), Wake and REM sit in the lower-right (highest δ -S/R, lowest α -S/R), and N1, N2 occupy intermediate positions on the same line. This inversion is the band-resolved expression of the well-known opposite trajectories of δ and α power across the sleep cycle. Critically, the S/R ratio is amplitude-invariant by construction, so the inversion lives in the temporal-information structure rather than in raw amplitude.

Figure 14: α vs δ S/R compass.



x = δ -band S/R ratio; y = α -band S/R ratio; one dot per (subject $\times$ channel $\times$ stage). Large X markers: stage centroids; ellipses: 95% covariance. N3 (upper-left) and Wake (lower-right) are at opposite ends of a near-linear stage manifold.

Discussion

N3 as the dominant multi-timescale information state

The strongest finding of this work is the exceptional elevation of all PID atoms during N3 slow-wave sleep. Figure 2 demonstrates that this is not a statistical artefact of pooling: all three atoms track sleep-stage identity in real time, rising sharply at each N3 block and returning to baseline at transitions to lighter or REM sleep. At the group level (Figure 3), N3 shows redundancy and synergy substantially above Wake, while N1 is the lowest stage on every magnitude atom. Figure 4 confirms the N3 elevation without exception across all 10 subjects and all six electrodes, ruling out an outlier-driven artefact and showing it is a reliable individual-level signature.

Both redundancy and synergy increase together in N3, consistent with the richly nested temporal environment of slow-wave sleep (co-occurring slow oscillations, spindles, and hippocampal sharp-wave ripples spanning seconds to minutes), though the PID analysis does not identify specific physiological generators. Wake and REM are indistinguishable on most PID magnitude atoms (BH-corrected $p > 0.05$, Figure 3), consistent with their shared profile of low-amplitude, thalamo-cortically desynchronised activity.

Within-stage variability dissociates N3 on R and U from S

The variability analysis (Figure 9) adds a qualitative dimension that the magnitude analysis cannot capture. N3 is simultaneously high-mean and low-CV on redundancy and total unique information, identifying it as a coherent, low-variance state on the linear-persistence atoms. By contrast N3 carries the highest CV for synergy, indicating that the joint predictive structure remains volatile window to window even where its mean is largest. The opposite-slope behaviour of S/R-ratio CV across stages (Wake, N1, N2 monotonic decline with lag; REM rising; N3 U-shaped) is inconsistent with pure correlation-decay, and a within-stage heterogeneity account (phasic versus tonic REM; slow-wave morphology in N3) is the simplest mechanism that matches the observed signs.

Synergy as a unique marker beyond local persistence

While all atoms are elevated in N3, only synergy exceeds the block-permutation null at the group level (Figure 5): several lag pairs in the intermediate region reach $p < 0.05$. Redundancy, though very high in absolute terms, falls within the range expected from the local within-window persistence preserved by the null, confirming that its elevation is largely explained by stronger second-scale autocorrelation during N3. Synergy, which requires *joint* information from two past timescales beyond what either provides alone, is not adequately captured by the null model.

Synergy here is a property of the discretised temporal representation under the MMI PID framework, not a direct physiological quantity; the concentration of significant synergy at intermediate lags is consistent with coupling at second-to-tens-of-seconds timescales but requires confirmation with stronger null models.

Stage differences across the lag grid: what the heatmaps show

The N3 – Wake contrast (Figure 8, row 1) presents the sharpest pattern. Both synergy and redundancy are uniformly and strongly elevated: BH-corrected paired t-tests reach *** ($p < 0.001$) across all 45 lag-pair cells for each atom, with no spatial concentration. This indicates that N3 elevates both temporal atoms as a broadband state property, independent of the specific (τ_1 , τ_2) combination tested. The S/R ratio difference is negative throughout (N3 has *lower* S/R ratio than Wake), indicating a trend toward greater redundancy-dominance in N3.

The REM – Wake contrast (row 2) reveals a selective profile. REM synergy is significantly elevated above Wake (***) in all 45 cells), but the magnitude is roughly 20% of the N3 – Wake effect. REM redundancy shows no significant cells after BH-FDR correction, indicating that REM and Wake are indistinguishable in their temporal redundancy structure across all tested timescales. Taken together, REM differs from Wake in synergy alone.

What the band-resolved pass adds

Two findings from the band-resolved section reshape the interpretation of the broadband pass. First, the broadband variability dissociation (Figure 9) is not a uniform band-mixing effect: only a subset of (band $\times$ atom) combinations actually carry stage information (Figures 10 and 11), and the rest contribute to the broadband mean without contributing to its stage modulation. Second, the α vs δ S/R inversion (Figure 14) is a clean, single-panel, amplitude-invariant phenomenon that broadband averaging entirely hides: δ and α S/R move in opposite directions across stages. The α -band inversion of the S/R ratio is the band-resolved expression of the opposite trajectories of α and δ power across the sleep cycle, but expressed in temporal-information structure rather than in raw amplitude.

Potential confounds and robustness checks

Delta-band amplitude. Three converging lines of evidence argue against an amplitude-confound account of the N3 elevation. First, per-window quantile binning renders discretisation amplitude-invariant within each window. Second, Figure 7 shows that within N3 the S/R ratio is flat across all observed delta-power levels: high-delta and low-delta N3 windows carry the same relative synergy-redundancy balance. Third, the Wake S/R-delta relationship is *negative*: drowsy pre-sleep epochs with incidentally elevated delta are *more* redundancy-dominated, the opposite of a straightforward amplitude confound.

Temporal autocorrelation. N3 has stronger autocorrelation due to slow oscillations, which could inflate PID through trivially persistent signals. The block-permutation test probes this directly: it preserves within-window structure but destroys all cross-window autocorrelation at the tested lag scales. Redundancy does *not* exceed the null, indicating its elevation is consistent with local autocorrelation. Synergy *does* exceed the null at several lag pairs.

Stage boundary contamination. Temporal triplets were retained only when every window in the triplet's span was scored as the same stage, preventing cross-boundary transitions from inflating within-stage statistics.

Coverage bias. Cross-stage comparisons were restricted to the subset of lag pairs present in every stage (common-lag correction), so per-window means are computed over identical timescale subsets for all stages.

Limitations and future directions

Stronger null models. The block-permutation null establishes genuine second-scale temporal ordering structure but does not rule out linear autoregressive explanations. AR-matched surrogates fitted per stage and subject, and IAAFT phase-randomised surrogates preserving spectral shape, are needed before drawing stronger inferential conclusions.

Representation dependency. All results are conditional on the parameterisation (1-s windows, $N_b = 4$ quantile bins, 1-30 s lag grid, MMI measure). Results should be cross-validated with alternative PID measures (e.g. BROJA; Bertschinger et al., 2014) and replicated under sensitivity analyses varying window length and binning resolution.

Per-channel block-perm dropped. The block-permutation null was computed only on the C3 composite to bound wall-time; per-channel maps would have added approximately 30 hours to compute.

Sparse joint at 1-s windows. 4 samples per joint-distribution cell on average ($N_b = 4$, 256 samples per window). Per-window PID estimates are noisier than at longer windows; the global PID matrix and the time-resolved median across lag pairs are far more reliable than any individual window estimate.

Band-resolved pass on 3 subjects. The band-resolved Results section is based on 18 (subject $\times$ channel) units per band; extending to all 10 subjects is a natural next step.

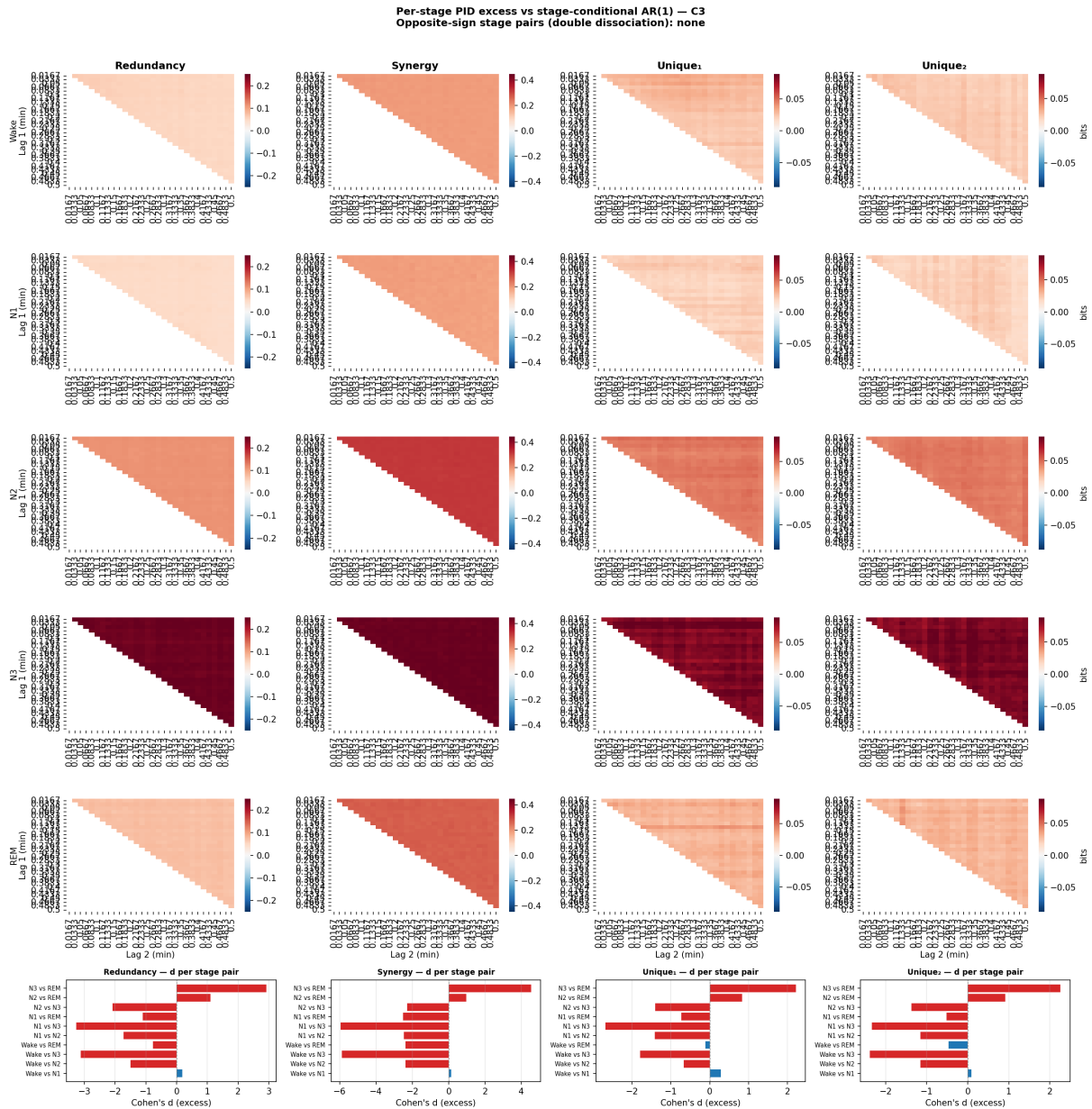
Conclusion

Temporal Partial Information Decomposition reveals clear and statistically robust stage-dependent structure in overnight EEG at the second timescale. N3 slow-wave sleep is distinguished by large, simultaneous increases in all information atoms, and by a low-variance signature on redundancy and total unique information that identifies it as a coherent multi-timescale state. Block-permutation testing identifies synergy as the PID atom that most consistently exceeds the local-persistence null, marking multi-lag joint predictive structure as a candidate distinguishing feature of cortical temporal organisation. The band-resolved analysis identifies an α vs δ inversion of the synergy-to-redundancy ratio across stages, an amplitude-invariant phenomenon that broadband averaging hides and that suggests information-decomposition measures track stage identity in ways complementary to spectral power.

Supplementary figures

S1: Per-stage PID excess vs stage-conditional AR(1). A direct test of whether stages exceed their own linear baseline in qualitatively different ways: for each stage we compute the lag-pair excess (Actual – stage-AR(1)) matrix per atom; below the heatmap grid is a strip showing Cohen's d of the per-window excess for every stage pair. Stage pairs whose d-vector has opposite signs across atoms (e.g. $d > 0$ for redundancy but $d < 0$ for synergy) are flagged as double-dissociation candidates. At the single-subject level shown here (sub-1, C3), the four atoms all show same-sign excess across stages, consistent with the entropy-rate scaling that makes the atoms covary in magnitude. The variability dissociation of Figure 9, with N3 low-variance on R / U but high-variance on S, is the more informative qualitative dissociation across stages.

Figure S1: Per-stage PID excess vs stage-conditional AR(1) (sub-1, C3, representative).



Rows: sleep stages. Columns: redundancy, synergy, unique₁, unique₂. Each cell is the $(\tau_1 \times \tau_2)$ excess matrix (Actual – stage-AR(1)) with diverging colourmap centred at zero; a per-atom shared $v_{\max}$ is used across stages. Bottom strip: Cohen's d of per-window excess per stage pair, one bar per atom.

Software and Reproducibility

All analyses were implemented in Python 3.10. Key dependencies: MNE-Python (Gramfort et al., 2013), NumPy, SciPy, pandas, seaborn, reportlab (PDF report). The PID atoms are computed in pure NumPy, verified against dit (James et al., 2018) to 10^{-15} .

Data are publicly available at <https://openneuro.org/datasets/ds005555>. Analysis scripts:

```
scripts/pid/eeg_sleep_compute.py # per-subject compute
scripts/pid/eeg_sleep_plot.py # per-subject plots
scripts/pid/eeg_sleep_group_figs.py # group stage comp, effect sizes, C3 block-perm
scripts/pid/eeg_sleep_extra_figs.py # subject consistency, electrode topo, delta
confound
scripts/pid/eeg_sleep_bands_paper_figs.py # band-resolved Results figures
scripts/utils/build_sleep_report_v2_pdf.py # this PDF
```

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